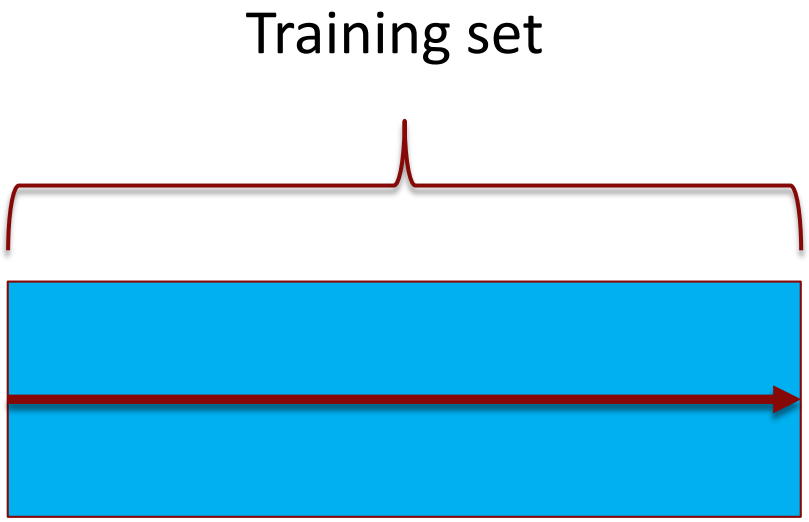
(Recap) Gradient Descent vs Stochastic Gradient Descent

- At each iteration, use all data points to calculate the gradient of the loss function

- At each iteration, randomly choose just a few of the data points and use only these to compute the gradient of the loss function

Epochs and Batches

What is an epoch?



An epoch is one pass through the full training set.

But this plays out differently for Gradient Descent vs *Stochastic* Gradient Descent.

An epoch in Gradient Descent

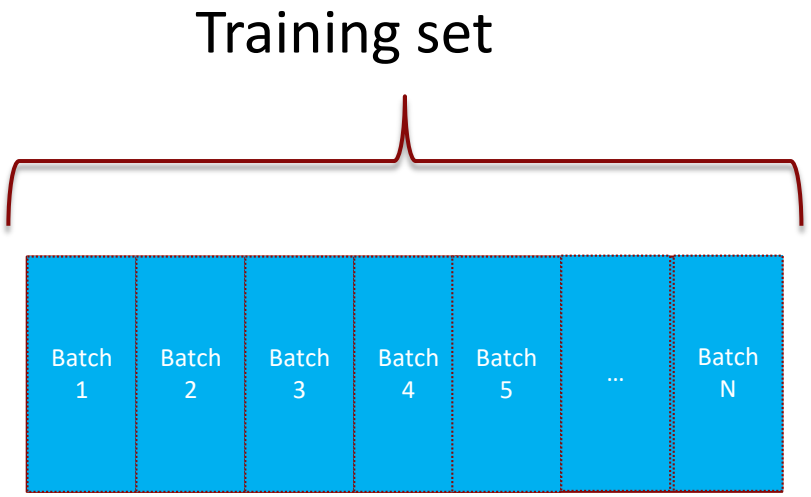


- We run every training sample through the network to get the predictions
- We calculate the gradient of the loss
- We update the parameters

$$w \leftarrow w - \alpha \frac{dLoss(w)}{dw}$$

This is done just once at the end of the epoch

An epoch in Stochastic Gradient Descent

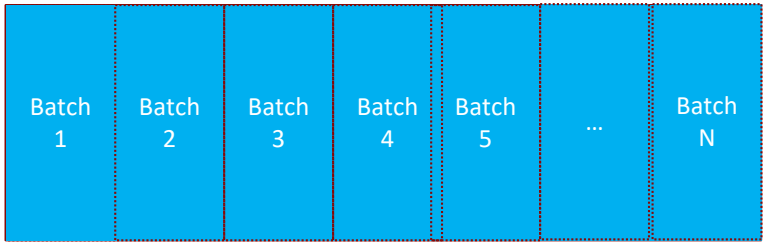


But when we do Stochastic Gradient Descent (SGD), we process the data in minibatches, one after the other*

*we will refer to minibatches as batches from now on for simplicity

An epoch in Stochastic Gradient Descent

Training set

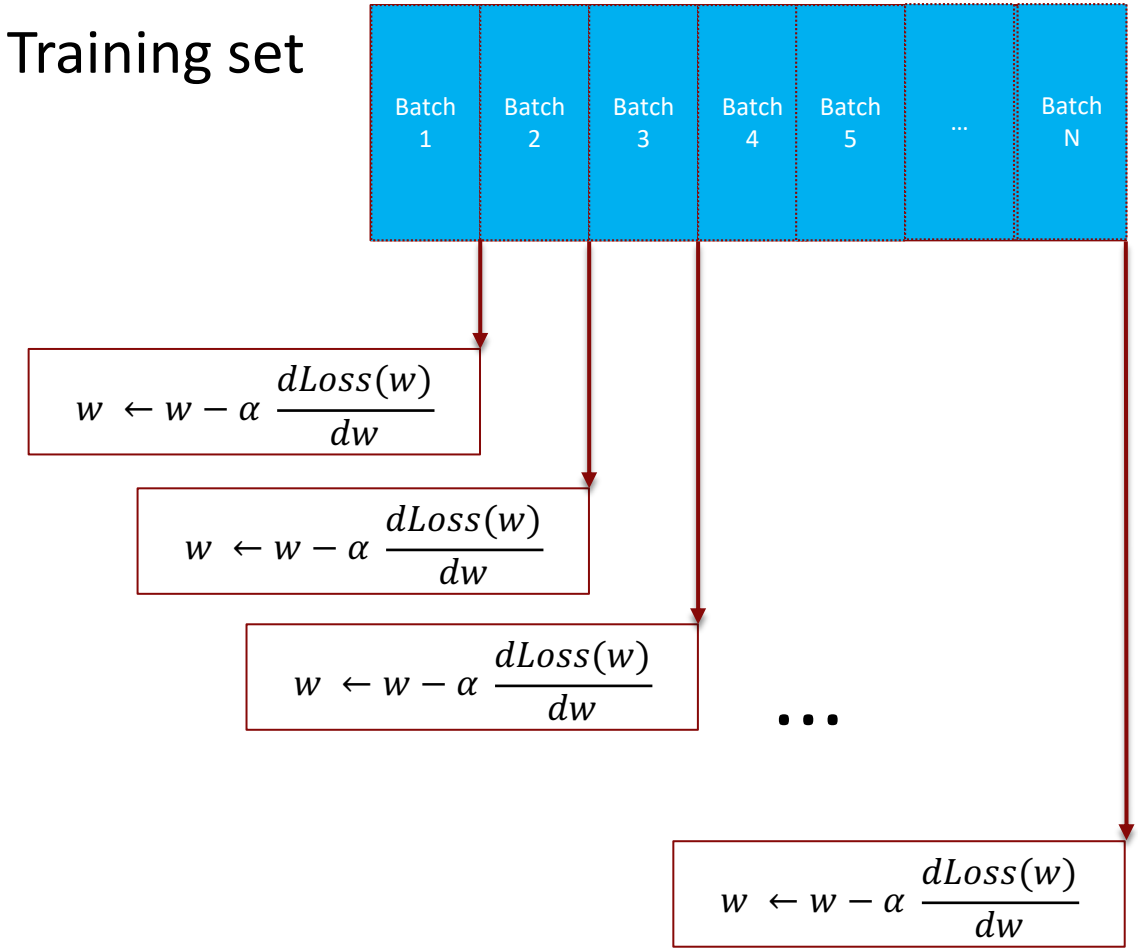


For each batch:

- *We run the training samples in that batch through the network to get predictions*
- *We calculate the gradient of the loss*
- *We update the parameters*

$$w \leftarrow w - \alpha \frac{dLoss(w)}{dw}$$

An epoch in Stochastic Gradient Descent



How many batches in an epoch when we do SGD?

of batches in one epoch = (Training set size / Batch size) rounded up

For Neural Heart Disease Model:
Training set size = 194
Batch size = 32
of batches in one epoch = (194/32) rounded up = 7

The first 6 batches have 32 samples each, and the 7th batch has the last 2 samples.
32 * 6 + 2 = 194

